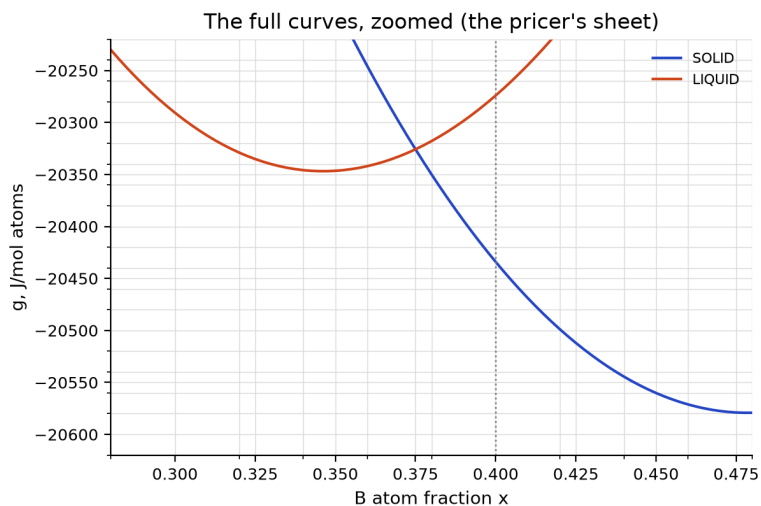
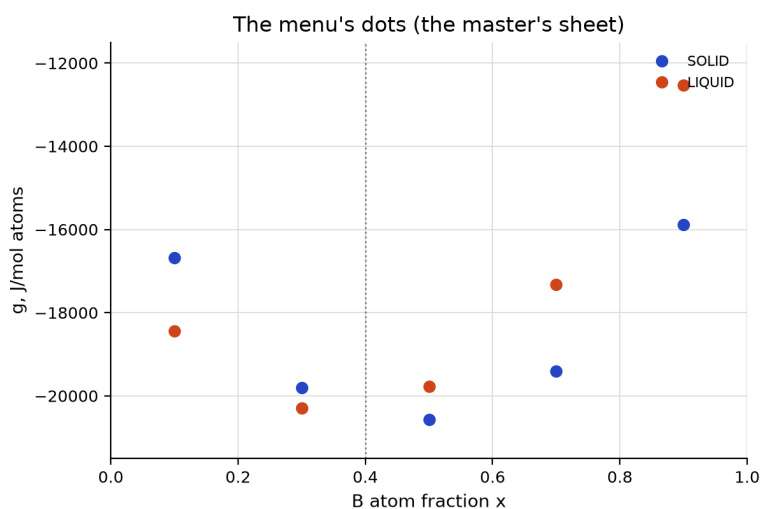


Advanced steps 07-18: the column-generation game

For two players or two teams. The **master** holds the dots and draws a line; the **pricer** holds the full curves and offers a state below the line, or says “none”. Then swap roles. You need this sheet, a transparent ruler (the “line”) and a pencil.

Rules.

1. The master marks the sample at $z = 0.40$, chooses the cheapest mixture of the dots on the master’s sheet (lever rule), and lays the ruler through the two used dots. The master announces the line’s heights at $x = 0$ and $x = 1$.
2. The pricer’s sheet is zoomed to x from 0.28 to 0.48. The pricer works out the line’s height at both edges, $\mu_A + \Delta\mu x$ (with $\Delta\mu = \mu_B - \mu_A$), draws the line there with the ruler, and looks for any point of the curves below it. If there is one, the pricer names its composition and phase model; the master adds it as a new dot.
3. Repeat until the pricer says “none”. Write down the ceiling (the master’s best mixture) after every round.
4. Debrief: how did you know when to stop? How far below the ruler was the deepest point the pricer found in the first round?



For the debrief. First round: the master's line has $\mu_A = -19873.9$ and $\mu_B = -21262.9$; the deepest point below it is SOLID near 0.45, -61.2 J/mol atoms. The true answer is -20473.12 .